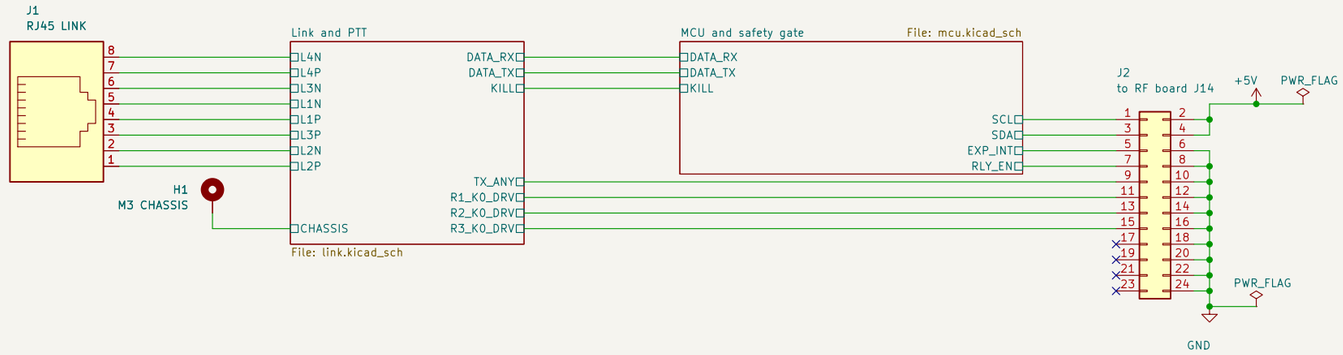


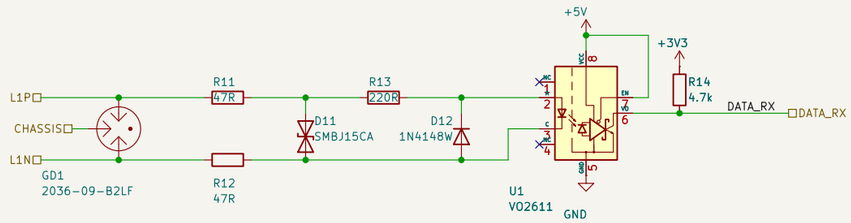
Controller board, unit 1 (architecture rev C, D−51, D−55 to D−58). Stacked above the RF board: J2 plugs onto RF board J14.
RJ45 LINK on the top wall: Cat6 #1 to the shack, T568B, four opto current loops (L1 data in, L2 PTT, L3 KILL, L4 data out).
Safety gate on the MCU sheet (D−59): supervisor with watchdog, fault latch and NOR gate drive RLY_EN (J2 pin 7).
RLY_EN has its 100k pull−down on the RF board: without the controller board all relays stay off.

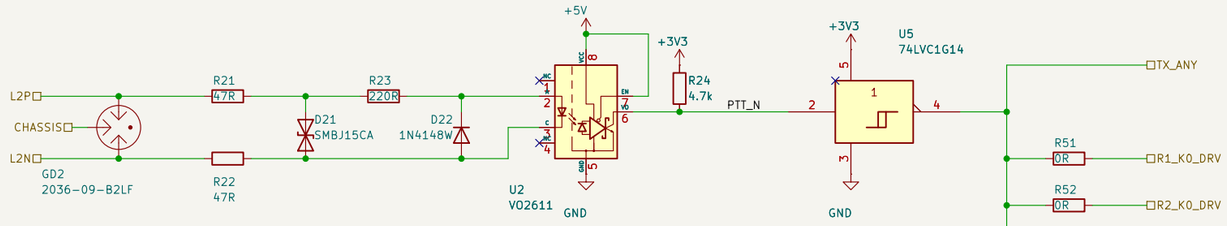
Stackup and size: to decide with the layout
J2 2x12 to RF board J14 (D−51, D−52). PTT to K0 in hardware (no MCU)
ESP32−C3−DevKit−RUST−2 (D−55), RJ45 LINK 4 opto current loops (D−56, D−57)
Design: Prins Doornekamp, PE1RGU
PE1RGU − Shack Relay Matrix

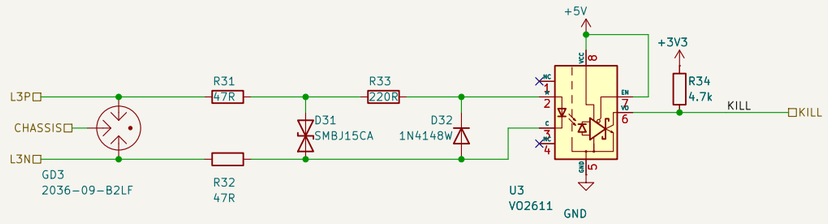
Sheet: /
File: controller−board.kicad_sch

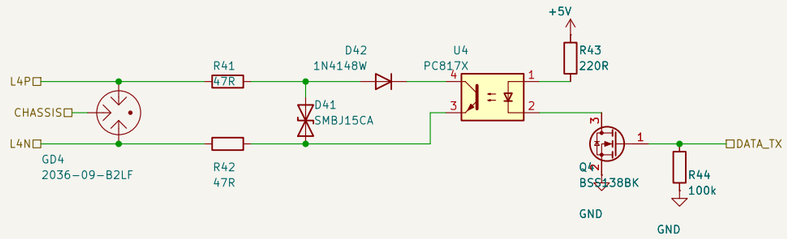
Title: Controller board: overview and connectors

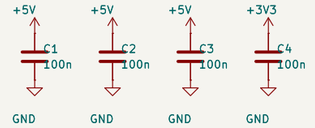
Size: A3	Date: 2026−09−28	**Rev: C**
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Link to the shack (D-56, D-57): Cat6 #1 UTP, four opto current loops, 10 mA each, powered from the shack (12 V, about 820R there).
T568B pairs, + wire first: pair 1 = pins 4/5 data in, pair 2 = 1/2 PTT, pair 3 = 3/6 KILL, pair 4 = 7/8 data out.
Entry per pair: GDT to CHASSIS (box earth), 47R MELF per wire, SMBJ15CA across the pair; then the opto.
Receive: loop current = LED on = output low. DATA_RX idle (mark) is low: the UART RX is inverted in firmware (ESP32-C3 line inverse).
PTT: FT-710 TX GND closes loop 2; PTT_N low; U5 gives TX_ANY and the K0 drives, in hardware, not through the MCU.
KILL: loop 3 closed through the shack KILL switch; open loop, cut cable or no shack supply = KILL high (to the safety gate).
Transmit: DATA_TX high (UART idle) = LED on = loop closed; MCU off = loop open, so the shack sees a break. 9600 baud (C-18).

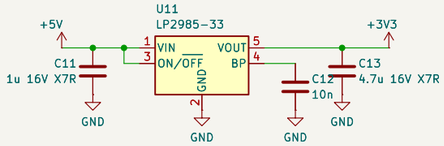
Stackup and size: to decide with the layout
J2 2x12 to RF board J14 (D-51, D-52). PTT to K0 in hardware (no MCU)
ESP32-C3-DevKit-RUST-2 (D-55), RJ45 LINK 4 opto current loops (D-56, D-57)
Design: Prins Doornekamp, PE1RGU

PE1RGU – Shack Relay Matrix

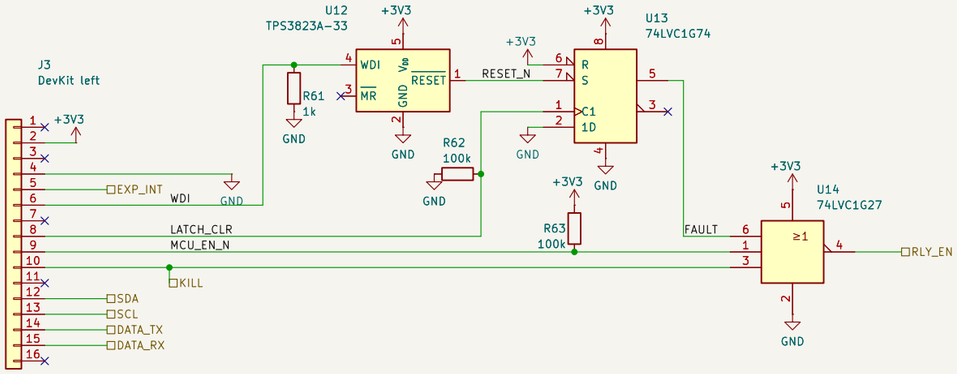
Sheet: /Link and PTT/
File: link.kicad_sch

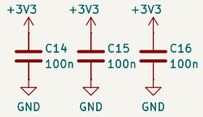
Title: Controller board: link to the shack and PTT

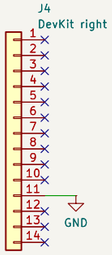
Size: A3	Date: 2026-09-28	**Rev: C**
KiCad E.D.A. 10.0.5		Id: 2/3



Safety gate (D-59)







MCU: ESP32-C3-DevKit-RUST-2 Rev 2.0 on two sockets (D-55). 3V3 pin from U11 (LDO); ENABLE low keeps the board buck off;
5V/VBUS not connected, so the PC can stay on USB (board diode D6). Wi-Fi and BLE off in firmware.
I2C: board bus SDA GPIO7, SCL GPIO8 (10k on the board; 2.2k on the RF board), SHTC3 0x70 and ICM-42670-P 0x68 on the board.
GPIO0 EXP_INT, GPIO1 WDI (kick at least every 0.9 s), GPIO3 LATCH_CLR (rising edge), GPIO4 MCU_EN_N (low = relays allowed),
GPIO5 KILL (read), GPIO20/21 UART 9600 baud. GPIO2, 8, 9 are strapping pins: GPIO2 and GPIO9 not used; GPIO8 is SCL. GPIO6 spare.
Safety gate (D-59): RLY_EN = NOR(KILL, MCU_EN_N, FAULT). FAULT is set by RESET_N: power-up, 3V3 below 2.83 to 3.00 V, or no WDI edge for 0.9 to 2.5 s.
After a reset the relays stay off until the MCU gives a LATCH_CLR edge. KILL is not latched. RLY_EN has a 100k pull-down on the RF board.
U12 RESET_N does not reset the MCU (DevKit pin 1 RST open); the ESP32-C3 has its own watchdogs. MR open: internal 90k pull-up.

Stackup and size: to decide with the layout
J2 2x12 to RF board J14 (D-51, D-52). PTT to K0 in hardware (no MCU)
ESP32-C3-DevKit-RUST-2 (D-55), RJ45 LINK 4 opto current loops (D-56, D-57)
Design: Prins Doornekamp, PE1RGU

PE1RGU – Shack Relay Matrix

Sheet: /MCU and safety gate/
File: mcu.kicad_sch

Title: Controller board: MCU, 3.3 V and safety gate

Size: A3	Date: 2026-09-28	**Rev: C**
KiCad E.D.A. 10.0.5		Id: 3/3